

AJAY VENKATESH

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Experience

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Designed a secure UI Console architecture using Midway SSO, OAuth, and HELIS to enforce least-privilege, role-based access; presented the solution to the team and received design approval.
- Built serverless infrastructure using AWS CDK to deploy API Gateway, Lambda (Java 21), DynamoDB, and S3 with cross-account access; implemented CI/CD pipeline for multi-stage deployments.
- Developed the frontend using React 18, integrating AWS UI components, Midway AuthN, and HELIS AuthZ with role-aware views and filtering; delivered a UI that was 40% faster and significantly more secure than older version.
- Implemented backend APIs using Java Coral, Smithy models, Dagger DI, and CBOR serialization; improved response time by 35% and ensured reliability through comprehensive unit and integration testing.

New York University

Aug 2024 – May 2025

Software Developer (On-Campus)

New York, US

- Contributed to an AI-driven Remote Patient Monitoring System, awarded \$100K by the National Science Foundation, USA, employing React Native, TypeScript, and Cognito authentication for real-time health monitoring.
- Developed an AI model that identifies food items through mobile images and predicts calorie, protein, and fat intake using YOLO + OCR based classification; integrated the model into the app workflow and stored results in the database.
- Built parallelized ETL pipelines in Python to transform wearable NoSQL data into structured SQL, reducing processing time by 60%, and visualized health trends using AWS QuickSight dashboards with secure row-level access.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Associate

- Built a distributed backend on Microsoft Azure for a health platform, integrating Microsoft Dynamics for secure data flow and using Azure services for scalability, fault tolerance, and event-driven processing.
- Led code reviews & mentored interns on design, TDD, and optimization, reducing bugs by 25% and accelerating CI/CD in an Agile setup.
- Developed a scalable B2B platform with modular services to replace legacy onboarding, integrating pipelines, email tracking, and BI reporting, improving automation by 70%.
- Optimized data pipeline by parallelizing Excel file imports (100K+ rows), cutting processing time from 5 hours to 1 and boosting efficiency by 80%.
- Designed and deployed a cognitive AI solution using OCR and NLP on Azure, integrating it with business workflows through development, ensuring scalability, security, and compliance with PwC delivery standards.
- Led project delivery across design, development, testing, and deployment, aligning with PwC's SDLC framework, coordinating with PMs, architects, and business stakeholders.
- Implemented RBAC and session validation for 5+ roles, enforcing least-privilege access and preventing unauthorized actions.

Projects

Flight Delay Prediction

- Built a scalable PySpark ML pipeline on HDFS to process and engineer features from over 7M flight records using SQL-based aggregation and anomaly detection.
- Performed extensive EDA to identify delay patterns and key predictors; trained and optimized Random Forest and Gradient Boosted Tree models to achieve **85%+ accuracy**.
- Converted model outputs into actionable insights, enabling data-driven scheduling and operational planning for delay reduction.

Skills

Programming Languages: C#, Python, Java, C++, JavaScript, SQL

Database: Dataverse, RDBMS, NoSQL

Cloud: Azure (AI Services, Azure AI Foundry, Cognitive Services, Logic Apps, Function Apps, Storage Blobs, Security Data Lakes), AWS (CDK, SDK, EC2, S3, Lambda, IAM, Q, SageMaker),

AI/ML: Neural Networks, Classification, Pattern Recognition, Data Extraction, RAG, MLOps, AI Solution Deployment, Responsible AI, Model Monitoring, Model Validation & Monitoring

Automation Platforms: Power Automate, Microsoft Copilot Studio, Azure AI Studio

Tools: Git, Docker, Jupyter, Visual Studio, Postman, Power BI, CI/CD Pipelines

Methodology: Agile, Scrum, SDLC Governance, Test-Driven Development (TDD)

Education

New York University | Master of Science in Computer Engineering (GPA – 3.9/4)

Aug 2024 – Aug 2026

Certifications

Azure Data Fundamentals

Power Platform Developer Associate

Power BI Data Analyst Associate

Power Platform Solution Architect Expert

Azure Data Scientist Associate

AWS Cloud Practitioner

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*) [Link]

An Aquila-optimized SVM classifier for Diabetes prediction (*IEEE*) [Link]